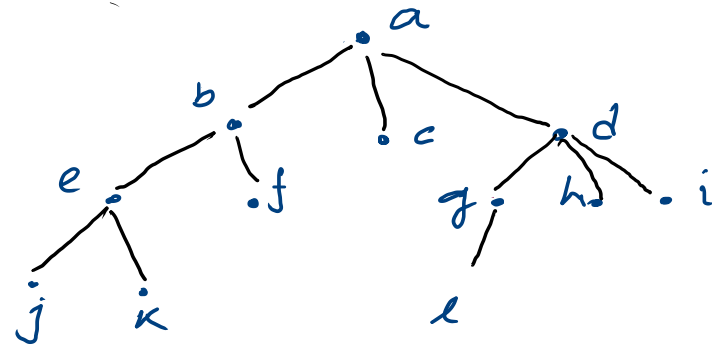


Chapter 11.3 Tree traversal

pre-order: Root, L → R

in-order: Leftmost, Root, L → R

post order: L → R, Root



Pre-order

a b c d
a b e f c d g h i
a b e j k f c d g h i

In-order

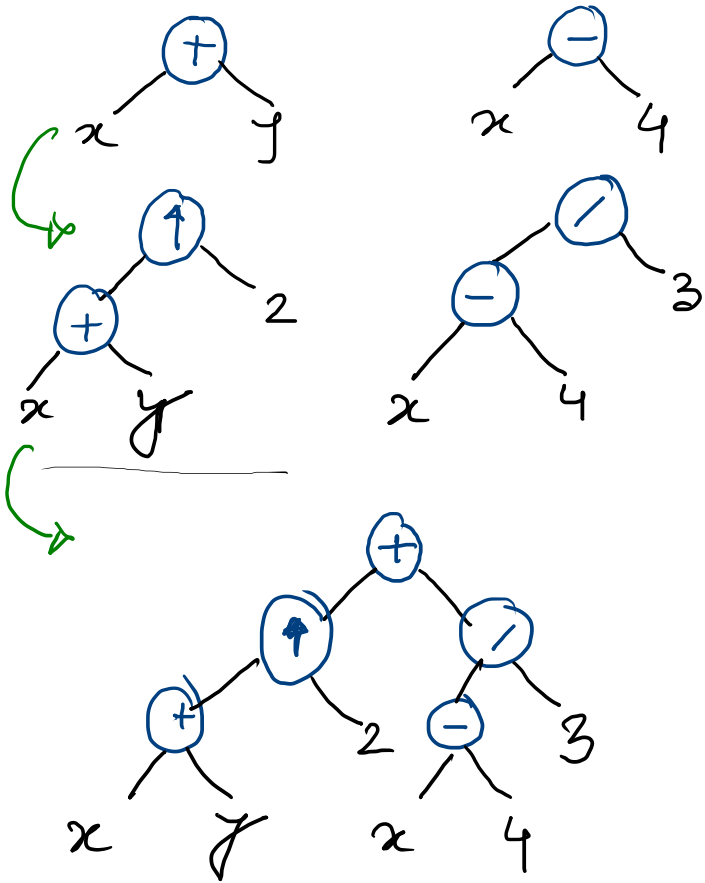
b a c d
e b f a c g d h i
j e k b f a c l g d h i

Post-order

b c d a
e f b c g h i d a
j k e f b c l g h i d a

*) Ordered rooted tree for expression

$$\rightarrow ((x+y)^2) + ((x-4)/3) =$$



- prefix (pre-order)

$$+ \uparrow /$$

$$\rightarrow + \uparrow + 2 / - 3$$

$$\rightarrow + \uparrow + x y 2 / - x 4 3$$

- infix (in-order)

$$\uparrow + /$$

$$\rightarrow + \uparrow 2 + - / 3$$

$$\rightarrow x + y \uparrow 2 + x - 4 / 3$$

- postfix (post-order)

$$\uparrow / +$$

$$\rightarrow + 2 \uparrow - 3 / +$$

$$\rightarrow x y + 2 \uparrow x 4 - 3 / +$$

④ Evaluating postfix, prefix

① Prefix : $R \rightarrow L$

+ - * 235 / $\uparrow \overline{2} \overline{3} \overline{4}$
 $\underbrace{\phantom{\uparrow \overline{2} \overline{3} \overline{4}}}_{2 \uparrow 3 = 8}$

$$+ - \cancel{2}35 / \color{green}{8}4$$

$$\underline{8\cancel{4} = 2}$$

$+ - \cancel{2} 3 5 2$
 $\quad \quad \quad \underline{\quad \quad \quad}$
 $\quad \quad \quad 2 \times 3 = 6$

$$\begin{array}{l} + \quad \underline{-652} \rightarrow + \quad \underline{-562} \rightarrow \underline{(5-6)+2=1} \\ \quad \quad \quad 6-5=1 \quad \quad \quad 5-6=-1 \quad \quad \quad \underline{\quad \quad \quad 1} \end{array}$$

$$+12 \quad \rightarrow \boxed{+ - 12} \text{ or } +(-1)2$$

$$1 + 2 = 3$$

2

$$(5-6) + 2$$

$$-1 + 2$$

$$(0 - 1) + 2$$

$$\begin{array}{r} + - 012 \\ \hline 0 - 1 = -1 \end{array}$$

$$+ - 12$$

⑧ Postfix : $L \rightarrow R$

$$7 \ 2 \ 3 \times - 4 \ 9 \ 3 / +$$

$$\underline{2 \times 3 = 6}$$

7 6 - 4 1 9 3 / +
7 - 6 = 1

$$\begin{array}{r} 1493/+ \\ \underline{14} = 1 \end{array}$$

$$\begin{array}{r} \underline{193} / + \\ 9/3 = 3 \end{array}$$

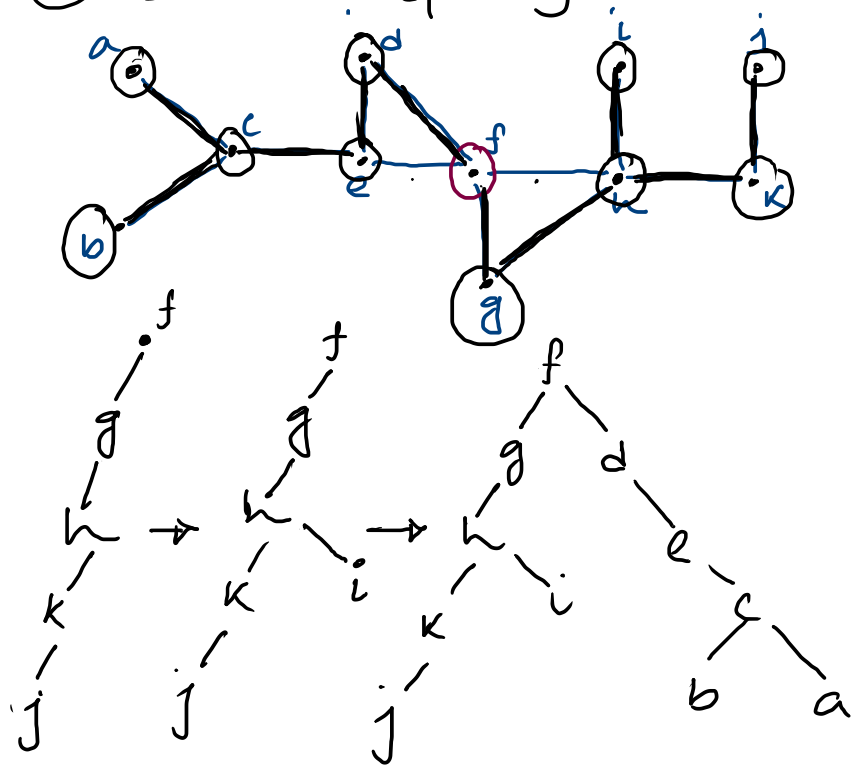
$$\begin{array}{r} 1 \overline{) 3} \\ 1 + 3 = 4 \end{array}$$

2

Chapter- 11.4 Spanning tree.

① **Spanning Tree**: the sub-graph of a graph G containing all the vertices of G .

② **DFS** $\rightarrow$ Depth first search



③ **BFS** - Breadth first search.

